Climate Justice or Climate Rhetoric?

Does International Multilateral Climate Finance
Reach the Most Vulnerable Countries?

Prithika Narayanan
Student Number: 63046

Submitted in partial fulfilment of the requirements for
ST451 Bayesian Machine Learning

Department of Statistics
London School of Economics and Political Science

May 2026

Contents

1. Introduction

International multilateral climate finance is governed by a single normative principle: countries bearing the least responsibility for historical emissions but facing the greatest climate risks should receive the most support. The 2009 Copenhagen pledge of USD 100 billion per year, renewed and raised to USD 300 billion at COP29, rests on this principle [UNFCCC, 2009]. This study asks whether the disbursement mechanism operates on it.

The question is not rhetorical. Existing empirical evidence points in different directions. Tennant et al. [2024] report a positive average relationship between vulnerability and absolute climate finance across 133 developing economies from 2000–2022, but find that institutional quality and conflict status significantly moderate the relationship. Lee et al. [2025] find the most vulnerable countries received relatively less World Bank adaptation finance over 2014–2023. Weiler et al. [2018] identify governance quality, trade links, and political relationships as the primary drivers of allocation rather than vulnerability itself. Scandurra & Thomas [2020] show that finance flows to bankable, large-scale projects, systematically favouring middle-income countries. These studies agree that something other than vulnerability drives allocation, but disagree on what that something is and whether vulnerability still has a residual positive effect once confounders are controlled.

The contribution here is to resolve the apparent tension through three analytical decisions, each of which changes the substantive conclusion. First, the outcome is **finance per capita** rather than absolute flows. India receives USD 42.9 billion in total, yet only USD 31 per person; absolute totals classify India as the best-served country in the dataset, while per-capita framing places it among the worst. The per-capita choice is not cosmetic — it changes the sign of the relationship, and therefore the answer to the justice question. Second, **Bayesian regression with prior sensitivity analysis, LOO-CV, Pareto-k diagnostics, and a Student-t robustness check** separates what the data imply from what the priors and likelihood assumed. Demonstrating that the negative coefficient survives every robustness check applied — but that its magnitude depends materially on the noise specification — is itself the substantive contribution in a literature characterised by point estimates without uncertainty quantification. Third, **GMM clustering paired with a hierarchical Bayesian model and a Frisch–Waugh–Lovell decomposition of vulnerability and institutional readiness** tests whether the vulnerability penalty operates through institutional capacity. The decomposition reveals that the two constructs correlate at $\rho = -0.887$ in this dataset and cannot be cleanly separated — a finding that, properly interpreted, sharpens the policy implication rather than dilutes it.

The central research question that organises the analysis is therefore:

> *Does per-capita multilateral climate finance flow toward vulnerability, as climate justice demands — or toward institutional capacity, as the disbursement architecture incentivises?*

The analysis proceeds in five stages: data integration and exploratory diagnostics (Sections 2, 4), methodology (Section 3), GMM typology and pooled Bayesian regression with robustness checks (Sections 5.1–5.6), the hierarchical and readiness-extended decomposition that identifies the operative mechanism (Sections 5.7–5.8), and policy implications (Section 6). Supporting figures are reported in the appendix.

2. Data

Three publicly available datasets are merged at the country level. The final dataset covers **152 countries** with **USD 761.9 billion** in recorded multilateral climate finance over 2000–2023.

Fan et al. [2025] provides project-level data from 18 multilateral institutions including the World Bank, ADB, AfDB, GCF, and GEF. Instrument type (loan, grant, guarantee, equity) is recorded per project, enabling analysis of the composition of finance rather than just its volume. Notre

Table 1: Data sources. Final merged dataset: 152 countries, USD 761.9bn total finance.

Dataset	Coverage	Variables
Climate Finance [Fan et al., 2025]	185 countries, 2000–2023	Finance, loans, grants, adaptation/mitigation
ND-GAIN Vulnerability [Notre Dame Global Adaptation Initiative, 2026]	192 countries, avg 2010–2023	5 vulnerability sub-dimensions, readiness
World Bank WDI [World Bank, 2026]	266 country codes	GDP per capita, population, GNI per capita

Dame Global Adaptation Initiative [2026] provides annual scores from 0 to 1 across six vulnerability sub-dimensions — food, water, health, ecosystems, infrastructure, habitat — averaged over 2010–2023. Five sub-dimensions are used as clustering features alongside the overall vulnerability score; mean vulnerability is 0.436 (std = 0.093, range = 0.255–0.645). ND-GAIN also publishes a **readiness** score capturing economic, governance, and social capacity to deploy climate finance effectively; this score is held in reserve for the decomposition analysis in Section 5.9, allowing the pooled and hierarchical models to be built without it and the institutional-capacity mechanism to be tested directly as a model extension. GNI per capita from the World Bank assigns income group classifications at 2023 thresholds. Distributional properties of the merged dataset appear in Appendix Figure 3.

3. Methodology

3.1 Preprocessing

Finance, GDP per capita, and population are log-transformed to address heavy right-skew. Per-capita finance is the outcome variable: the -0.48 correlation between log population and log finance means that without this adjustment, populous countries appear better served than they are. All features are standardised to zero mean and unit variance so that clustering treats each dimension equally and regression coefficients are directly comparable in standard-deviation units.

3.2 Gaussian Mixture Model Clustering

GMM is preferred over k-means for two reasons: countries lie on a continuum of vulnerability and finance, so probabilistic soft assignments better reflect the underlying structure than hard partitions; and GMM's full covariance structure accommodates clusters of different shapes and sizes, which matters when small-island and large-population vulnerable countries are expected to occupy different regions of feature space. Models are fitted for $K = 2$ to 8 (`covariance_type='full'`, `n_init=15`). BIC is minimised at $K = 2$; silhouette score is maximised at $K = 4$ (silhouette = 0.240). The $K = 4$ solution is selected because it maps onto a 2×2 (vulnerability $\times$ finance) framework that makes the climate justice claim testable at the level of country typology — the $K = 2$ solution collapses the theoretically central distinction between high-vulnerability-low-finance countries and high-vulnerability-high-finance countries into a single cluster.

3.3 Bayesian Linear Regression

The outcome is modelled as:

$$\log_{10}(\text{finance pc}_i) = \alpha + \beta_1 \tilde{v}_i + \beta_2 \widetilde{\log \text{GDP}}_i + \beta_3 \widetilde{\log \text{pop}}_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2). \tag{1}$$

Two pooled models are fitted using the No-U-Turn Sampler (4 chains, 2,000 draws, 1,000 tuning iterations). **Model 1 (Informative)**: $\beta_{\text{vuln}} \sim \mathcal{N}(0.5, 1)$ encodes the climate justice hypothesis; $\beta_{\text{GDP}} \sim \mathcal{N}(-0.5, 1)$; $\sigma \sim \text{HalfNormal}(1)$. **Model 2 (Sceptical)**: $\beta_{\text{vuln}} \sim \mathcal{N}(0, 0.5)$ encodes no directional prior expectation. The Bayesian approach is chosen not for mechanical reasons but because it returns a full posterior distribution over each coefficient, enabling direct probability statements ("$P(\beta_{\text{vuln}} < 0) = 98.5\%$") and quantifying prior sensitivity explicitly. Models are compared using PSIS-LOO-CV [Vehtari et al., 2017] and WAIC; Pareto-k diagnostics flag influential observations.

3.4 Hierarchical Bayesian Model

The pooled model treats all 152 countries as exchangeable. Where this assumption is violated — as the residual analysis confirms (low-income mean residual -0.163, upper-middle $+0.084$) — partial pooling across income groups is introduced via:

$$\mu_\beta \sim \mathcal{N}(0, 1), \quad \sigma_\beta \sim \text{HalfNormal}(0.5), \quad \beta^{(g)}_{\text{vuln}} \sim \mathcal{N}(\mu_\beta, \sigma_\beta).$$

Each income group g receives its own vulnerability coefficient drawn from a common hyperprior. The structure is later extended in Section 5.9 to decompose the vulnerability coefficient into a component absorbed by institutional readiness and a residual.

4. Exploratory Data Analysis

Four findings from the exploratory analysis motivate the formal modelling that follows. Each appears with a brief supporting statistic in text; full distributions and correlation structures are presented in Appendix A.

Finding 1 — Vulnerability and finance are not positively correlated. The scatter of vulnerability against total climate finance (Figure 1) shows no upward trend at either raw or log scale. Countries receiving the most finance — India, China, Turkey, Germany — sit in the 0.30–0.45 vulnerability range. Countries above 0.55 cluster at the bottom of the finance axis. The correlation between log finance and vulnerability is just 0.07. This is the central empirical puzzle the rest of the study resolves: the data show no evidence of the positive vulnerability–finance relationship that climate justice predicts.

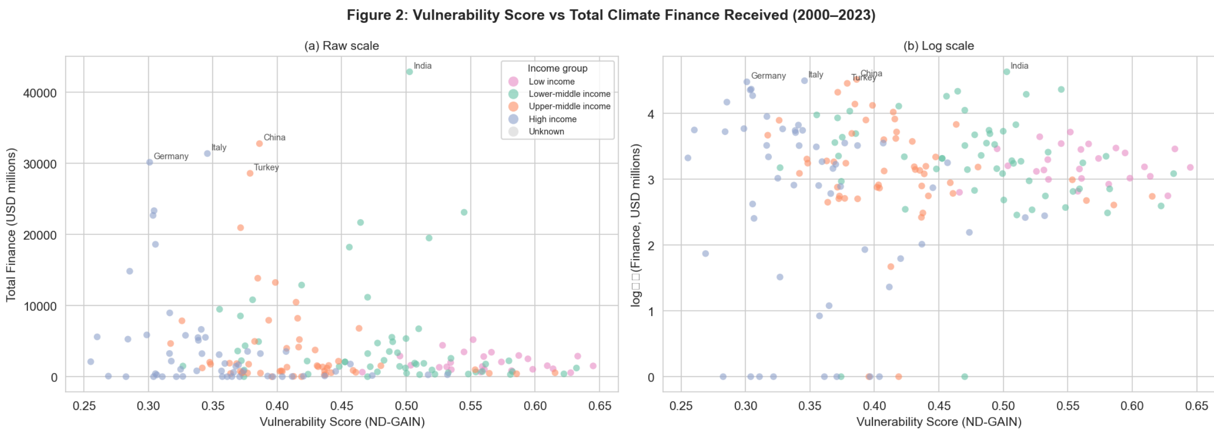

Figure 1: Vulnerability score vs total climate finance received (raw and log scale), coloured by income group. No positive relationship is visible at either scale. The absence of an upward slope — and the clustering of high-vulnerability countries at the bottom of the finance distribution — is the central empirical puzzle this study resolves.

Finding 2 — Volume and instrument inequality compound. Mitigation finance falls sharply from the least-vulnerable quartile (Q1) to the most-vulnerable (Q4); adaptation finance is comparable to mitigation only in Q4, but only because both reach their lowest absolute levels there (Appendix Figure 4). The GDP–vulnerability correlation of -0.82 means the poorest countries are systematically the most exposed, so the quartile pattern is also a wealth pattern. The most vulnerable countries receive the least of every type of finance.

Finding 3 — Per-capita framing is essential. The log population–log finance correlation of -0.48 means absolute totals systematically misclassify the most populous vulnerable nations as well-financed. This is why India appears as the dataset's best-served country in absolute terms but is among the nine worst-served per capita.

Finding 4 — Instrument composition compounds allocation. Q4 receives approximately 50% of its finance as repayable loans versus 89% for Q1, but the ratio obscures scale: Q4's mean grant volume (USD 618M) is an order of magnitude smaller than Q1's mean loan volume (USD 5,584M). A higher grant proportion of a much smaller total is not adequate compensation, and the effective terms facing vulnerable countries are therefore worse than the headline ratio suggests. The full correlation structure supporting these four findings is reported in Appendix Figure 5.

5. Results

5.1 GMM Clustering

Models were fitted for $K = 2$ through $K = 8$. BIC is minimised at $K = 2$ while silhouette score peaks at $K = 4$ (silhouette = 0.240); the disagreement reflects a genuine tension between parsimony and structural detail. The $K = 4$ solution is chosen because $K = 2$ collapses the theoretically central distinction between high-vulnerability-high-finance and high-vulnerability-low-finance countries into a single cluster, erasing the typological contrast the subsequent analysis requires. The full BIC and silhouette curves appear in Appendix Figure 6. The selected $K = 4$ solution achieves mean cluster assignment probability = 0.983, with only 2 uncertain assignments — the clustering is decisive at the country level and not a compromise between competing fits.

Table 2: GMM cluster profiles ($K = 4$).

Cluster	n	Mean Vuln.	Mean Finance	Dominant Income
C1: Low Vuln | High Finance	33	0.328	USD 6,609M	High income
C2: Low Vuln | Low Finance	17	0.375	USD 9M	High income
C3: High Vuln | High Finance	62	0.402	USD 5,366M	Upper-middle
C4: High Vuln | Low Finance	40	0.522	USD 4,495M	Low/Lower-middle

C3 is the cluster that most clearly disproves a purely climatic allocation story. India, China, Brazil, and Indonesia are highly vulnerable yet attract substantial finance — they possess the capacity to develop the large-scale bankable projects that multilateral institutions require. If the allocation mechanism were climatic, C3 would not exist; these countries would either be routed to the high-finance-low-vulnerability profile (C1, which they do not fit) or excluded from high finance entirely. The fact that they sit in a distinct high-vulnerability-high-finance cluster indicates the allocation mechanism is institutional, not climatic. C4 is where the system fails: 40 countries with the highest mean vulnerability (0.522) and the least institutional access. Within C4, per-capita finance spans from USD 18 (Zimbabwe) to USD 2,281 (Samoa) — a two-order-of-magnitude range within a single vulnerability cluster, driven almost entirely by population size. The C4 composition is consistent with the descriptive literature on adaptation finance: small-island states and sub-Saharan conflict-affected countries dominate the most-vulnerable-least-financed group.

5.2 The Climate Justice Gap

Countries in the top vulnerability quartile that also receive bottom-quartile per-capita finance number **nine**: Sudan (vulnerability 0.610, USD 33), DRC (0.566, USD 36), Ethiopia (0.552, USD 44), Yemen (0.526, USD 36), Zimbabwe (0.510, USD 18), Myanmar (0.508, USD 33), Angola (0.507, USD 57), India (0.503, USD 31), Nigeria (0.489, USD 26). Mean per-capita finance across these nine countries: **USD 1.32 per person per year** — approximately the cost of a bottle of water, distributed annually across 2.01 billion of the world's most climate-exposed people. This is the headline statistic the policy simulation in Section 6.5 quantifies against proportional-allocation counterfactuals.

5.3 Bayesian Regression: Prior Predictive Check and Posterior Results

The prior predictive check (Appendix Figure 7) confirms the priors are weakly informative over the observed data range. The posterior ends at $\beta_{\text{vuln}} = -0.247$ despite the informative prior placing mass at +0.5 — a shift of −0.784 units against the direction of the prior, which is stronger evidence of data dominance than simply fitting a model with sceptical priors would provide. The OLS and Bayesian posterior means coincide closely (OLS = −0.254, Bayesian = −0.247). The Bayesian R^2 posterior mean is 0.144 (94% HDI: [0.114, 0.166]; full posterior in Appendix Figure 8). The OLS point estimate of 0.167 sits at the upper edge of the HDI — the frequentist figure slightly overstates stable explained variance.

Table 3: Regression model comparison. †OLS point estimate. ‡Bayesian posterior mean $R^2 = 0.144$ (94% HDI: [0.114, 0.166]); see Appendix Figure 8.

Model	MSE	R^2	Uncertainty Quantification
Null model	0.716	0.000	None
OLS regression	0.596	$0.167^\dagger$	Point estimate only
Bayesian regression	0.596	$0.144^\ddagger$	Full posterior + 94% HDI

Table 4: Bayesian posterior summary. Rows 1–4: informative priors. Row 5: sceptical prior.

Parameter	Mean	SD	94% HDI	$\hat{R}$	ESS
β_{vuln} (Informative)	−0.247	0.113	[−0.454, −0.032]	1.0	4,600
$\beta_{\text{GDP per capita}}$	−0.309	0.115	[−0.519, −0.089]	1.0	4,320
$\beta_{\text{population}}$	−0.338	0.067	[−0.465, −0.211]	1.0	8,355
σ	+0.788	0.046	[+0.704, +0.876]	1.0	7,647
β_{vuln} (Sceptical)	−0.230	0.107	[−0.431, −0.030]	1.0	5,238

All $\hat{R} = 1.0$, effective sample sizes exceed 4,000, no divergences — the MCMC is well-mixed and the posterior is reliably characterised. $P(\beta_{\text{vuln}} > 0) = 1.5\%$ under the informative prior and 1.4% under the sceptical prior. The negative GDP coefficient ($\beta_{\text{GDP}} = -0.309$) indicates that, conditional on vulnerability and population, lower-GDP countries receive more per-capita finance — a partial-derivative reading that does not contradict the unconditional pattern (large absolute flows concentrate in middle- and high-income economies) because vulnerability and GDP per capita are themselves correlated at $\rho = -0.82$. The full posterior densities under both priors, together with the forest plot and posterior predictive check, are reported in Appendix Figures 9, 10, and 11; the principal takeaway is that the prior sensitivity shift for β_{vuln} is 0.017, less than 7% of the posterior standard deviation, and all three coefficients are negative under both specifications.

5.4 LOO-CV, WAIC, and Influential Observations

The ELPD difference between the informative and sceptical models is −0.14 (SE = 17.49): predictively indistinguishable. WAIC corroborates (difference < 1 SE). Pareto-k diagnostics (Appendix Figure 13) show all 152 countries with $k < 0.5$ — the entire distribution sits below both the moderate (0.5) and high-influence (0.7) thresholds. The negative vulnerability coefficient is therefore not an artefact of any single outlier: not Samoa's extreme per-capita finance, not India's extreme population, not any combination. The finding is robust to leave-one-out perturbation of the dataset.

5.5 Robustness to Likelihood Specification: Gaussian vs Student-t

The Gaussian posterior predictive check (Appendix Figure 11) reveals two features the linear-Gaussian model cannot accommodate: a pronounced spike at $\log_{10}(\text{finance pc}) \approx 0$, corresponding to countries that receive essentially zero per-capita multilateral finance, and a slight underprediction in the upper tail. To test whether the spike reflects genuine heavy-tailed noise rather than zero-inflation, the pooled regression is refit with a Student-t likelihood, $\varepsilon_i \sim t_\nu(0, \sigma)$, learning the degrees of freedom ν from the data under an Exponential(1/30) prior. A small posterior ν would indicate genuinely fat tails; a large ν would indicate the Gaussian was adequate.

The data drive ν to a posterior mean of 1.1 (94% HDI: $[0.8, 1.4]$), with $P(\nu < 10) = 100\%$. The

LOO comparison strongly favours Student-t: ΔELPD = +43.69 against a standard error of 19.77. Taken at face value, this is decisive evidence for heavy-tailed noise. **Taken with care, it is something more interesting**: $\nu \approx 1.1$ is essentially a Cauchy distribution, which has no defined mean or variance. A model whose tails are this heavy is one that can never be "surprised" by an extreme observation, which inflates predictive log-likelihood mechanically rather than because the noise structure is genuinely Cauchy. The Gaussian–Student-t comparison shown in Appendix Figure 12 confirms this reading visually: both posterior predictive distributions miss the observed spike at zero finance, but the Student-t spreads enough mass into the lower tail to absorb that spike statistically without modelling it explicitly. The Student-t is, in effect, papering over zero-inflation by exploiting infinite-variance tails.

This matters for the headline coefficient. Under the Gaussian likelihood, $\beta_{\text{vuln}} = -0.230$ (94% HDI: $[-0.431, -0.030]$); under Student-t, the estimate attenuates to $\beta_{\text{vuln}} = -0.077$ (94% HDI: $[-0.160, +0.008]$), $P(\beta_{\text{vuln}} < 0) = 96.1\%$. **Neither model is well specified** — the Gaussian under-weights the zero-spike, the Student-t over-compensates with Cauchy-like tails — and the properly specified zero-inflated mixture is identified as the priority future extension in Section 8.2. What *is* robust is the direction: β_{vuln} is negative under Gaussian, Student-t, both prior choices, the hierarchical model, the C4 subset, and leave-one-out perturbation. Its magnitude depends on the noise assumption and is best read as bracketed by -0.08 and -0.25. The policy simulation in Section 6.4 flags both bounds: the Student-t-implied redistribution gap is approximately a third smaller than the Gaussian estimate.

5.6 Calibration and Residual Analysis

Within-group residuals by income class — -0.163 for low-income, $+0.020$ for lower-middle, $+0.084$ for upper-middle, -0.043 for high income — violate the pooled model's exchangeability assumption. The full calibration plot, showing C4 countries clustering systematically below the perfect-calibration line, appears in Appendix Figure 14. The interpretation is direct: the pooled model predicts more finance for C4 countries than they actually receive, and vulnerability is the variable that explains the gap. A single global vulnerability coefficient cannot adequately describe low-income and upper-middle income countries simultaneously, which motivates the hierarchical model in Section 5.8.

5.7 C4-Only Regression: Between- vs Within-Cluster Effects

The full-sample coefficient of -0.247 averages across all 152 countries, which raises the question of whether it reflects a structural difference between clusters or a genuine within-cluster relationship. Re-running the same Bayesian model restricted to the 40 C4 countries yields $\beta_{\text{vuln}} = -0.042$ (94% HDI: $[-0.145, +0.069]$) with $P(\beta < 0) = 77\%$. The HDI crosses zero, so a null within-cluster effect cannot be ruled out at 94% credibility. At $n = 40$, the test is genuinely underpowered and the wide HDI is partly a small-sample artefact. The substantive interpretation, however, is clear: the direction remains consistent with the full sample but the magnitude is much smaller. The primary mechanism driving the pooled coefficient is structural — the sorting of institutionally weak, highly vulnerable countries into C4 — rather than a within-cluster vulnerability gradient. This finding is important for how the policy implications are framed: it is not that every additional unit of vulnerability within a cluster reduces finance, but that the cluster membership itself — which is determined by institutional capacity and exposure jointly — is what drives disbursement. The side-by-side comparison of the full-sample and C4-only posteriors appears in Appendix Figure 15.

5.8 Hierarchical Bayesian Model

The residual analysis revealed systematic within-group bias in the pooled model. The hierarchical model addresses this by allowing the vulnerability coefficient to vary by income group via partial pooling. All four group-specific coefficients are negative (Table 5), and the LOO comparison yields an ELPD difference of -0.29 (SE = 17.50): predictive performance is equivalent to the pooled model. The hierarchical model's value is therefore inferential rather than predictive — it establishes that the negative vulnerability effect is not an artefact of aggregating structurally distinct groups, which is a stronger robustness claim than the pooled model alone can support.

Table 5: Hierarchical model: income-group vulnerability coefficients. Hyperprior mean $\mu_\beta = -0.269$ (94% HDI: $[-0.579, +0.010]$). Cross-group SD $\sigma_\beta = 0.176$.

Income Group	β Mean	94% HDI	$P(\beta < 0)$	n
Low income	-0.305	$[-0.567, -0.025]$	98.6%	15
Lower-middle income	-0.195	$[-0.457, +0.050]$	92.8%	44
Upper-middle income	-0.250	$[-0.574, +0.075]$	93.8%	44
High income	-0.333	$[-0.639, -0.031]$	98.7%	49
Pooled (reference)	-0.247	$[-0.454, -0.032]$	98.5%	152

The hyperprior mean $\mu_\beta = -0.269$ has a 94% HDI of $[-0.579, +0.010]$ that just crosses zero at the upper end. This does not undermine the finding: the hyperprior represents the population average across groups, and its HDI width reflects genuine cross-group heterogeneity ($\sigma_\beta = 0.176$) rather than uncertainty about direction. The more informative quantities are the group-specific estimates, two of which exclude zero decisively (low income: $P(\beta < 0) = 98.6\%$; high income: $P(\beta < 0) = 98.7\%$). The negative vulnerability penalty appears consistently across all four income groups. The full visualisation of the four group-specific posteriors and the population hyperprior is reported in Appendix Figure 16.

5.9 Controlling for Institutional Readiness: The Mechanism Test

The hierarchical model confirms the vulnerability penalty is not an income-group artefact, but does not identify *why* the penalty exists. The C3 cluster (India, China, Brazil, Indonesia) demonstrated descriptively that vulnerability does not mechanically block access: highly vulnerable countries with institutional capacity do receive substantial finance. This suggests the operative mechanism is **institutional readiness** — the capacity to develop bankable projects, absorb loan conditionality, and satisfy multilateral procurement requirements. The ND-GAIN readiness score, which aggregates economic, governance, and social capacity indicators, provides a direct measure — but the variable correlates with vulnerability at $\rho = -0.887$, near-perfect collinearity, which means any mediation analysis must explicitly handle the shared component.

Extending the hierarchical model with raw readiness yields $\beta_{\text{readiness}} = +0.267$ (94% HDI: $[-0.067, +0.629]$, $P > 0 = 92.5\%$). The vulnerability coefficients shrink toward zero in all four income groups by an average of **46.5%**: low-income moves from -0.305 to -0.215; high-income from -0.328 to -0.177. At face value this is consistent with institutional capacity as the operative mechanism. But the $\rho = -0.887$ correlation means the shrinkage could reflect either genuine mediation or pure relabelling of shared variance — a further test is required. Following Frisch–Waugh–Lovell logic, readiness is residualised against vulnerability ($\tilde{r}_i^{\perp} = r_i - \hat{\alpha} - \hat{\gamma}\, v_i$), isolating the component of readiness *not* explained by vulnerability. Refitting with $\tilde{r}_i^{\perp}$ in place of raw readiness yields a striking result: the orthogonal readiness coefficient is $+0.119$ (94% HDI: $[-0.057, +0.274]$, $P > 0 = 91.3\%$) — smaller than the raw version — and the vulnerability coefficients **grow larger in magnitude**, not smaller. Low-income moves from -0.305 to -0.450; the average change is -42.3%, away from zero. The full decomposition is reported in Appendix Table 6; the raw-readiness shrinkage pattern — the visualisation that initially appeared to support the institutional-capacity hypothesis before the orthogonal test was run — is shown in Appendix Figure 17.

The two results decompose a single reality. The 46.5% shrinkage under raw readiness reflects almost entirely the **shared variance** between vulnerability and institutional capacity (+88.8% of the total shrinkage); only -42.3% is attributable to readiness-specific variation, with a sign reversal. The interpretation is not that readiness fails as a mechanism, nor that the institutional-capacity story is wrong — it is that **vulnerability and institutional capacity are empirically inseparable in this dataset**. The climate finance architecture does not discriminate against "vulnerability" as distinct from "low institutional capacity" because in the 152 countries observed, those two conditions co-occur. The negative pooled and hierarchical vulnerability coefficient is not overturned — it is a robust data-level fact — but its interpretive content sharpens: it should be read as "the joint construct of vulnerability-and-institutional-weakness is systematically associated with less per-capita finance," not as "donors penalise vulnerability per se." Most importantly for policy, the inseparability strengthens rather than weakens the practical implication: reforming the disbursement architecture to reach the most vulnerable countries is indistinguishable from investing in the institutional capacity of those same countries. The two are not competing strategies but one.

6. Discussion

6.1 Per-Capita Framing Reconciles, Rather than Contradicts, Prior Literature

Tennant et al. [2024] report a positive average relationship between vulnerability and absolute climate finance. The result here is the opposite — in per-capita terms. These are not contradictory: Tennant et al. ask whether vulnerable governments receive more finance; this study asks whether the populations in those countries receive more per person. Both can be true simultaneously, and in the most populous vulnerable countries they are. India, Bangladesh, Nigeria, and Ethiopia each rank high on absolute receipts and low on per-capita receipts. A government that collects USD 42.9 billion while its citizens receive USD 31 each has not been served by the climate justice principle — it has been counted toward a donor's headline target. The per-capita framing is not a methodological preference; it is the operational test of whether climate finance functions as climate *justice*. A pledge denominated in absolute flows can be fulfilled while the populations the pledge was designed to help remain chronically under-financed, and the historical record of the USD 100 billion commitment supports exactly this reading.

A reasonable counter-argument is that per-capita framing itself misrepresents reality: climate finance projects have fixed costs and economies of scale, so USD 5bn invested in Bangladeshi grid modernisation may help 170 million people more than USD 30 per person scattered thinly. This argument has force for some project types but not all. Adaptation finance — seawalls, drought-resistant agriculture, early warning systems, community-level health infrastructure — is deployed locally and scales with exposed population; economies of scale operate only at the project level, not at the aggregate allocation level. More importantly, the economies-of-scale argument cannot rescue the allocation actually observed: nine countries housing 2.01 billion vulnerable people receive USD 1.32 per person per year. No plausible economies-of-scale story makes that figure consistent with climate justice, because the shortfall is not concentrated in low-population countries where scale could conceivably justify it — India and Nigeria, two of the most populous countries on Earth, are in the gap group.

6.2 The Structural Mechanism: Vulnerability and Institutional Capacity Are Inseparable

Three results jointly characterise the mechanism. First, C3 rules out a simple story in which vulnerability mechanically blocks access: vulnerable countries with capacity do get finance. Second, the C4-only regression ($\beta = -0.042$, HDI crosses zero) rules out that within-C4 countries show a strong internal vulnerability gradient — the differences among the most vulnerable are not what drives allocation within that group. Third, and most informatively, the readiness decomposition (Section 5.9) reveals that vulnerability and institutional readiness correlate at $\rho = -0.887$ and cannot be cleanly separated as distinct causal channels. The 46.5% shrinkage of vulnerability coefficients under raw readiness reflects almost entirely the variance these two constructs share; the orthogonalised analysis, which isolates readiness-specific variation, attributes only a minority of the original effect to readiness alone and produces vulnerability coefficients that grow rather than shrink. The honest reading is not that institutional capacity is or is not the mechanism — the data cannot tell those apart. The honest reading is that **in the climate finance system as observed, vulnerability and institutional incapacity co-occur to such a degree that the disbursement architecture treats them as a single condition**. The Bayesian R^2 of 0.144 (HDI [0.114, 0.166]) completes the picture: vulnerability, income, and population explain only a seventh of the variation in per-capita finance. The remaining 83–86% belongs to the political, procedural, and relational variables Weiler et al. [2018] identified — political ties, trade relationships, procurement compatibility, donor strategic preferences. The pattern is consistent with an account in which multilateral institutions face procedural constraints on what they can disburse, and vulnerable populations in institutionally weak states fail the procedural test twice: once because their governments cannot generate bankable project pipelines, and once because the same governments cannot absorb the conditionality that accompanies the finance that does flow.

6.3 The Instrument Composition Problem

Q4 receives approximately 50% of its finance as repayable loans versus 89% for Q1, but the headline ratio obscures the absolute scale. Q4's mean grant volume (USD 618M) is an order of magnitude smaller than Q1's mean loan volume (USD 5,584M). A higher grant proportion of a much smaller total is not adequate compensation. For countries where external debt already exceeds 60% of GNI, the climate finance designed to help them cope simultaneously increases the debt burden constraining their capacity to respond. The instrument composition problem is not a separate issue from the allocation problem; they are the same problem observed at different levels. A country that cannot generate bankable projects also cannot absorb loan-heavy finance, so the composition of what does flow to C4 is shaped by the same institutional-capacity constraint that drives the quantity.

6.4 Policy Implications: The Architecture, Not the Headline Figure, Is the Constraint

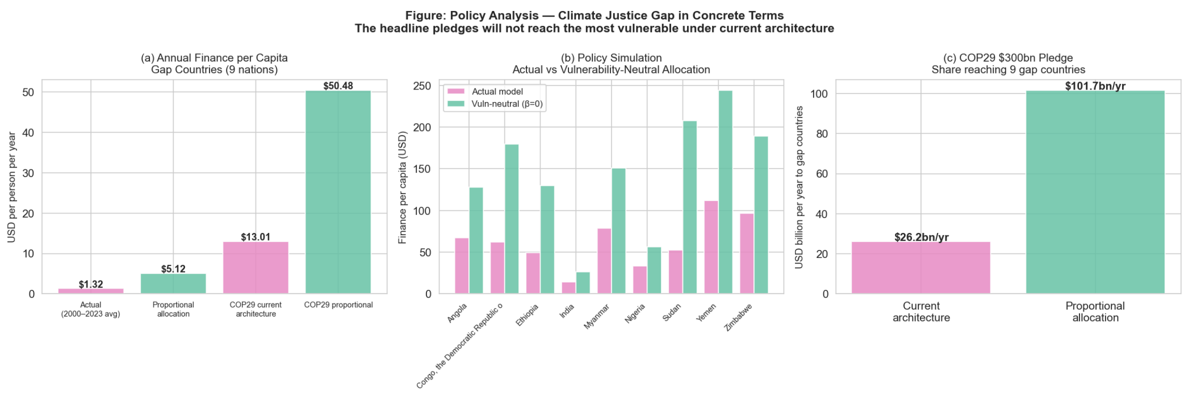

Figure 2: Policy analysis (Gaussian-likelihood counterfactual). Panel (a): per-capita finance, observed (USD 1.32) vs proportional (USD 5.12) and COP29 scenarios. Panel (b): setting $\beta_{\text{vuln}} = 0$ increases gap-country finance by 125%. Panel (c): COP29 reaches USD 26.2bn vs USD 101.7bn under current vs proportional architecture. Implied annual gap is USD 75.5bn under Gaussian; under Student-t (Section 5.5) approximately a third smaller.

The nine gap countries house 2.01 billion people and received USD 1.32 per person per year over 2000–2023. Under a vulnerability-proportional counterfactual using the Gaussian estimate, the regression-implied allocation is USD 5.12 per person — a shortfall of USD 3.80 per person annually. Setting $\beta_{\text{vuln}} = 0$ in the posterior simulation corresponds to a **125%** increase in modelled per-capita finance to gap countries. Extended to the COP29 commitment of USD 300 billion per year, this yields USD 26.2bn flowing to gap countries under current allocation patterns (USD 13.01/person) versus USD 101.7bn under the proportional counterfactual (USD 50.48/person), implying a USD 75.5bn annual structural gap — roughly a quarter of the COP29 commitment. **These are**

model-based counterfactuals contingent on the Gaussian likelihood; under the Student-t likelihood (Section 5.5), where the vulnerability coefficient attenuates to -0.077, the implied gap shrinks to roughly USD 50bn annually. The redistribution magnitude is therefore best read as the bracket [USD 50bn, USD 75bn], depending on the noise specification. The direction of the policy implication is robust to either model; the precise magnitude is not.

The inseparability of vulnerability and institutional capacity (Section 5.9) changes what "redirection" would require in practice. If the disbursement architecture is treating vulnerability and institutional weakness as a single condition — and the $\rho = -0.887$ correlation suggests it cannot do otherwise with the data available — then a substantial share of the modelled shortfall cannot be closed by allocation policy alone, because the receiving institutional infrastructure does not currently exist to absorb more finance even if it were offered. The policy simulation in Figure 2 quantifies the upper bound of what allocation reform alone could achieve under the assumption that the institutional constraint could be relaxed. Raising the headline pledge without addressing that constraint reproduces the pattern documented here at larger scale. The number is not the binding constraint; the procedural architecture is.

7. Conclusions

The multilateral climate finance system does not allocate according to climate justice principles. More vulnerable countries receive less per-capita finance. The **direction** of this effect survives every robustness check applied: both prior specifications ($P(\beta_{\text{vuln}} > 0) < 2\%$), all four income groups in the hierarchical model, leave-one-out cross-validation (no influential observations beyond the Pareto-k threshold), and the Student-t likelihood ($P(\beta_{\text{vuln}} < 0) = 96.1\%$). The **magnitude** is less robust: the Gaussian point estimate of -0.230 attenuates to -0.077 under the better-fitting Student-t, and the true value is most plausibly bracketed between the two. The Student-t fit also reveals a specification problem the Gaussian was hiding — a spike of countries receiving zero multilateral finance — that a zero-inflated model would address more honestly. The Frisch–Waugh–Lovell decomposition of vulnerability and institutional readiness, in turn, reveals that the two constructs correlate at $\rho = -0.887$ and cannot be cleanly separated. The 46.5% shrinkage observed when raw readiness is added to the hierarchical model reflects shared variance, not a clean mediation; under orthogonalised readiness, the vulnerability coefficients grow rather than shrink. The honest reading is that vulnerability and institutional weakness are empirically inseparable in this dataset, and the disbursement architecture treats them as a single condition.

The evidence points to three architectural changes whose direction is robust to the modelling choices that affect magnitude. Allocation targets should be set in per-capita terms, not in absolute flows. Grants should replace loans for the most debt-burdened recipients. Fragile states require upstream investment in project development capacity before they can access the finance that nominally exists for them. The USD 300 billion COP29 pledge will reproduce the pattern documented here unless the disbursement mechanism changes. The headline figure is not the binding constraint; the procedural architecture is.

8. Limitations and Future Work

Four caveats. First, the Gaussian–Student-t robustness check (Section 5.5) revealed neither likelihood is fully appropriate; the headline coefficient is best read as bracketed by -0.077 and -0.230 rather than as a precise point estimate. Second, the readiness-extended models do not separately identify vulnerability and institutional capacity as causal channels in cross-sectional data — $\rho = -0.887$ precludes this. The inseparability is a property of this dataset and time window (multilateral flows, 2000–2023); a different sample — bilateral aid, post-2025 multilateral flows, or country-level panel data — might exhibit weaker collinearity and permit the two channels to be separated. The hierarchical model also omits regional clustering, conflict, and governance indicators [Tennant et al., 2024]. Third, ND-GAIN scores carry measurement noise; classical attenuation means reported magnitudes are likely floors. Fourth, only multilateral public finance is covered.

The priority future extension is a **zero-inflated Bayesian model** of the form $y_i \sim (1-\pi_i)\delta_0 + \pi_i\mathcal{N}(\mu_i, \sigma^2)$, which would address the zero-spike directly rather than absorbing it through Cauchy-like tails; secondary extensions include bilateral flows, a Paris-Agreement temporal split, and governance/conflict controls.

References

Fan, S., Wang, C., Zhong, H., Dong, X., & An, K. (2025). International Public Multilateral Climate Finance Dataset from 2000 to 2023. *Figshare*. https://doi.org/10.6084/m9.figshare.28171535

Lee, N., Matthews, S., & Reid, J. (2025). Does World Bank Climate Adaptation Finance Go to the Most Vulnerable Countries? *Center for Global Development*.

Notre Dame Global Adaptation Initiative. (2026). *ND-GAIN Country Index*. University of Notre Dame. https://gain.nd.edu

Scandurra, G., & Thomas, A. (2020). Does climate finance reduce vulnerability in Small Island Developing States? *Journal of Cleaner Production*, 271, 122574.

Tennant, E., Gilmore, E., Schroeder, H., & Waitz, I. A. (2024). Impact of climate vulnerability on the allocation of climate finance: global evidence. *Science of the Total Environment*, 920, 171004.

UNFCCC. (2009). *Copenhagen Accord*. Decision 2/CP.15.

Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5), 1413–1432.

Weiler, F., Klöck, C., & Dornan, M. (2018). Vulnerability, good governance, or donor interests? The allocation of aid for climate change adaptation. *World Development*, 104, 65–77.

World Bank. (2026). *World Development Indicators*. https://databank.worldbank.org

Appendix

A. Exploratory Data Analysis — Supporting Figures

This appendix presents the exploratory diagnostics referenced in Section 4. Together they establish the four findings that motivate the formal modelling: the absence of a positive vulnerability–finance relationship (Figure 1, in the body); the distributional asymmetry in finance allocation (Figures 3–4); and the correlation structure that underwrites the per-capita reframing (Figure 5). Figure 6 reports the GMM model selection diagnostics underlying the choice of $K = 4$ in Section 5.1.

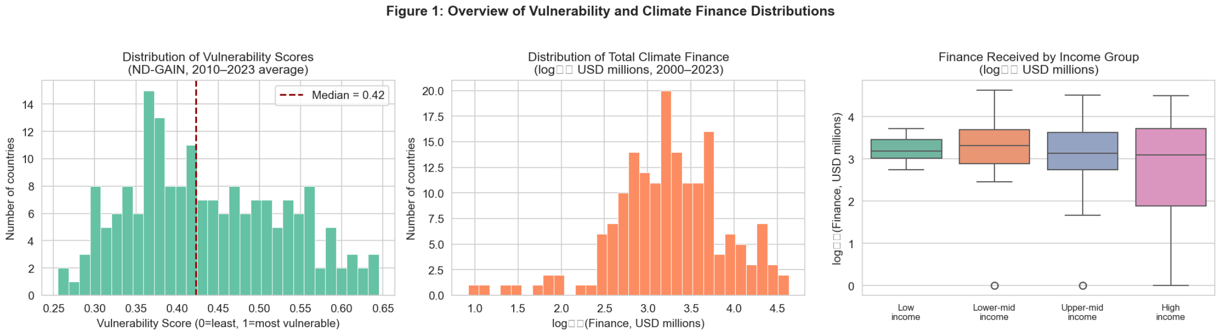

Figure 3: Distributions of vulnerability scores and climate finance. Vulnerability is approximately bell-shaped (median = 0.42) with a right tail above 0.55 dominated by low-income nations. Finance is highly concentrated even on a log scale. High-income countries show wider spread with more extreme outliers driven by loan-based finance — indicating that instrument type, not just volume, is a dimension of inequality.

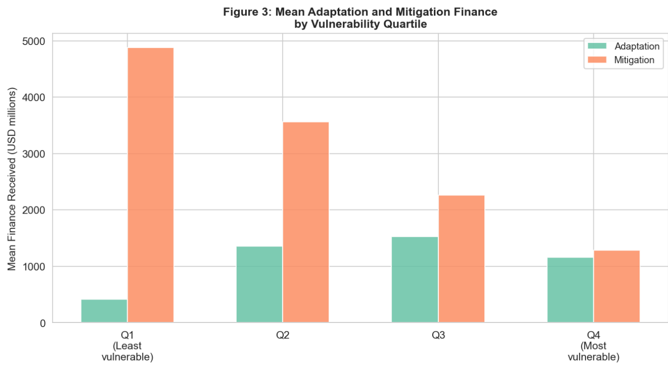

Figure 4: Mean adaptation and mitigation finance by vulnerability quartile. Mitigation dominates and declines from Q1 to Q4. Q4 is the only quartile where adaptation and mitigation are comparable — but both are at their lowest absolute levels there. The most vulnerable countries receive the least of every type of finance.

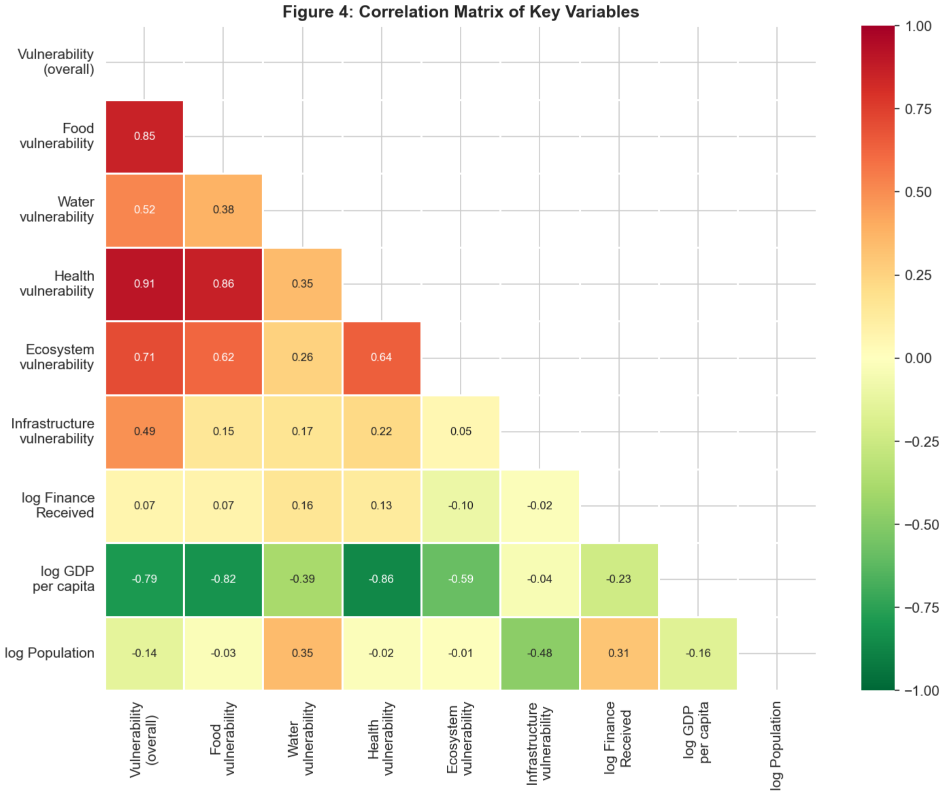

Figure 5: Correlation matrix. Three values anchor the analysis: vulnerability–log finance = 0.07 (near zero); log GDP–vulnerability = −0.82 (the poorest are most exposed); log population–log finance = −0.48 (per-capita framing changes who is well-served). Together they establish that the system does not function as climate justice predicts.

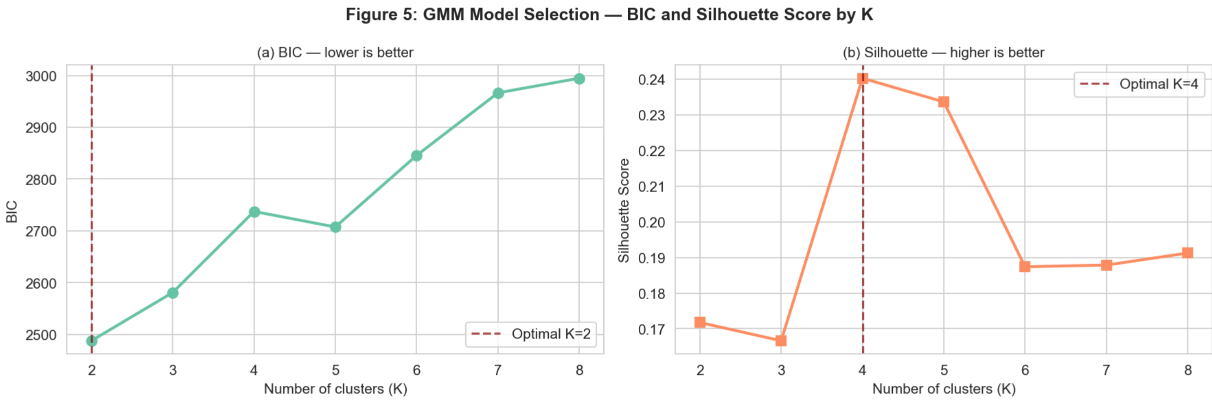

Figure 6: GMM model selection: BIC and silhouette scores for $K = 2$ to $K = 8$. BIC favours $K = 2$; silhouette peaks at $K = 4$. The $K = 4$ solution is chosen on substantive rather than purely statistical grounds: $K = 2$ collapses the theoretically central distinction between high-vulnerability-high-finance and high-vulnerability-low-finance country typologies into a single cluster.

B. Bayesian Diagnostics and Posterior Visualisations

This appendix presents the diagnostics and posterior visualisations supporting Sections 5.3–5.6. Figures 7–8 report the prior predictive check and the Bayesian R^2 posterior — the diagnostics establishing that the priors are weakly informative and that the explained variance carries genuine uncertainty. Figures 9–10 show the posterior densities and forest plot under both informative and sceptical priors, demonstrating that prior choice does not affect the substantive conclusions. Figures 11–12 are the central diagnostic for the model specification discussion: the Gaussian PPC reveals the zero-spike misfit, and the Gaussian–Student-t comparison shows that Student-t fits the bulk of the data better but absorbs the zero-spike via Cauchy-like tails rather than modelling it explicitly. Figures 13–14 report Pareto-k and calibration diagnostics: the former rules out single-observation influence on the posterior; the latter motivates the hierarchical model by showing systematic income-group bias in the pooled fit.

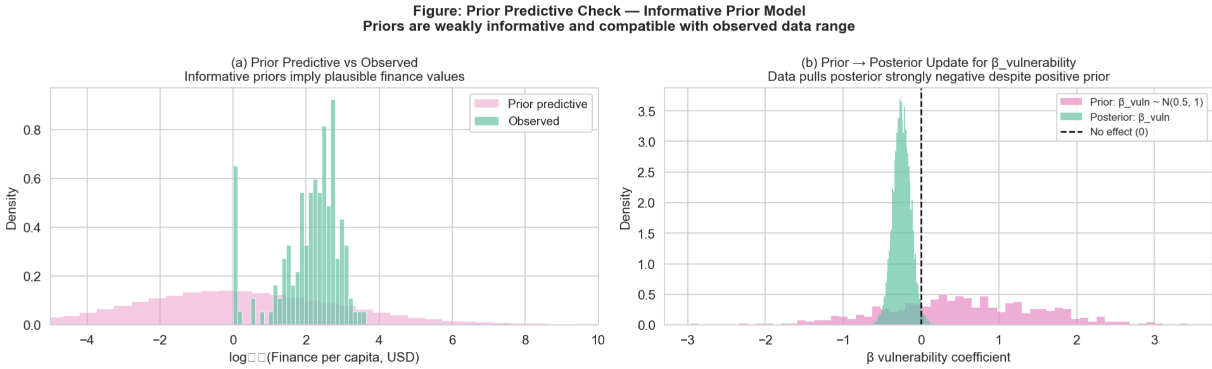

Figure 7: Prior predictive check. Panel (a): the prior predictive distribution is broad and compatible with the observed data range — the priors are weakly informative. Panel (b): the informative prior on β_{vuln} starts at $+0.5$ (climate justice hypothesis); the posterior ends at -0.247. A shift of -0.784 units against a positively-biased prior is strong evidence the result is driven by the data rather than the prior.

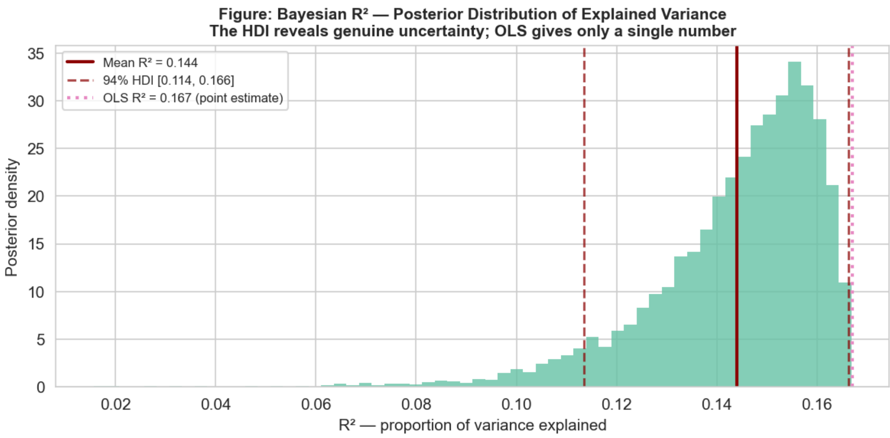

Figure 8: Bayesian R^2 posterior distribution (mean = 0.144, HDI [0.114, 0.166]). The OLS point estimate of 0.167 sits at the upper edge — the frequentist figure slightly overstates stable explained variance. The HDI also conveys the finding's substantive weight: three theoretically central predictors explain only 14–17% of variation. The remaining 83–86% reflects project pipeline availability, donor political preferences, and institutional compatibility — not climate need.

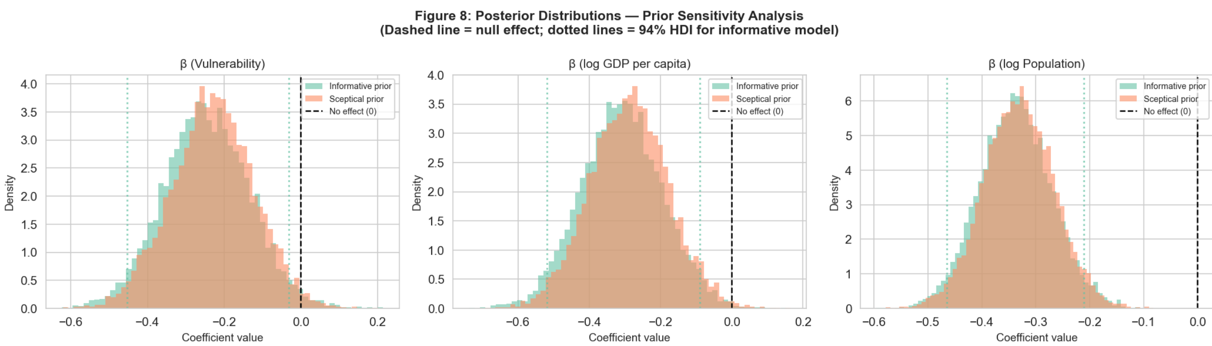

Figure 9: Posterior distributions under informative and sceptical priors. All three coefficients are negative under both specifications. The near-perfect overlap of the two posteriors — the prior sensitivity shift for β_{vuln} is 0.017, less than 7% of the posterior SD — means that starting from the climate justice hypothesis or from complete neutrality yields the same conclusion. The data dominate the prior entirely.

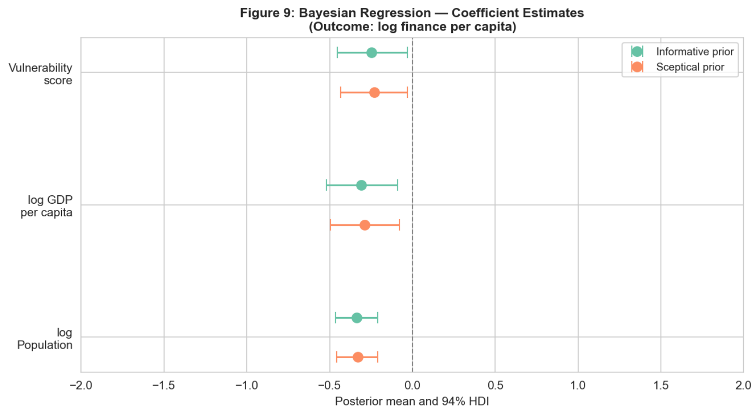

Figure 10: Forest plot: posterior means and 94% HDIs. All three coefficients are negative; the vulnerability HDI just excludes zero. The near-identical positions of the informative and sceptical estimates reinforce that prior choice is irrelevant to the substantive conclusion.

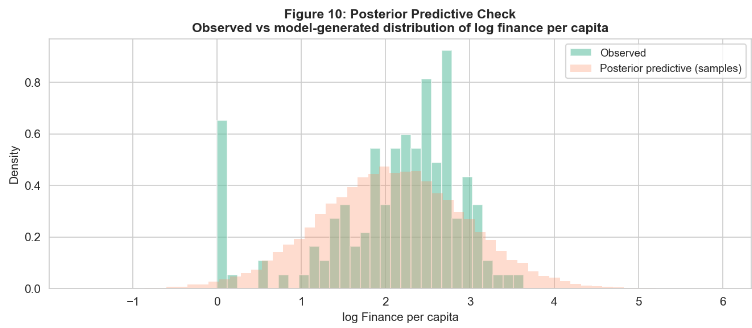

Figure 11: Posterior predictive check under Gaussian likelihood. Good fit across the bulk of the distribution but with two visible misfits: a pronounced observed-data spike at $\log_{10}$(finance pc) ≈ 0 that the Gaussian undershoots — the zero-inflation problem — and slight underprediction in the upper tail. Section 5.5 tests whether the misfit is heavy-tail noise or zero-inflation; Figure 12 shows the Student-t comparison.

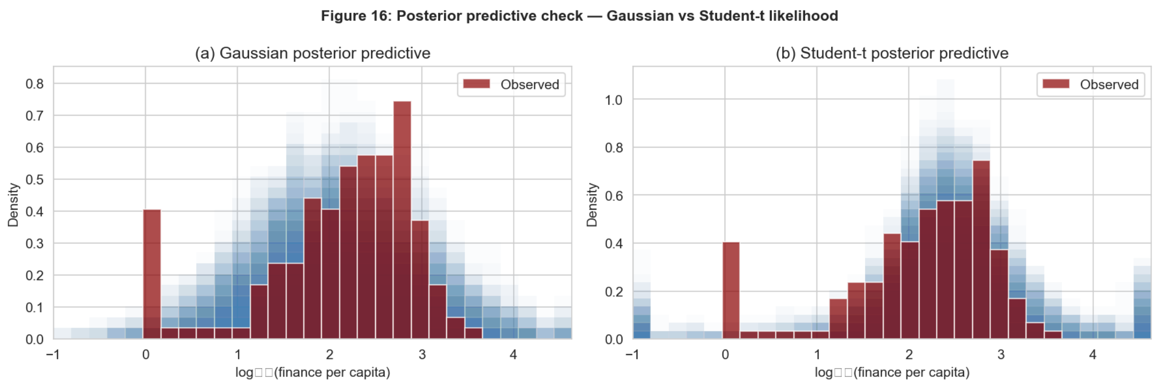

Figure 12: Posterior predictive comparison: Gaussian (left) vs Student-t (right) likelihoods, both clipped to the data range. The observed-data spike at $\log_{10}$(finance pc) ≈ 0 is visible in both panels and captured by neither: the Gaussian undershoots it, and the Student-t produces enough mass in the lower tail to absorb it statistically (via $\nu \approx 1.1$, near-Cauchy tails) but does not model it explicitly. The Student-t fits the central mass slightly better than the Gaussian and is preferred by LOO-CV (ΔELPD $= +43.69$), but the near-Cauchy tails are best read as a symptom of zero-inflation rather than evidence of genuinely fat-tailed noise. The properly specified model is zero-inflated and is identified as the priority future extension in Section 8.2.

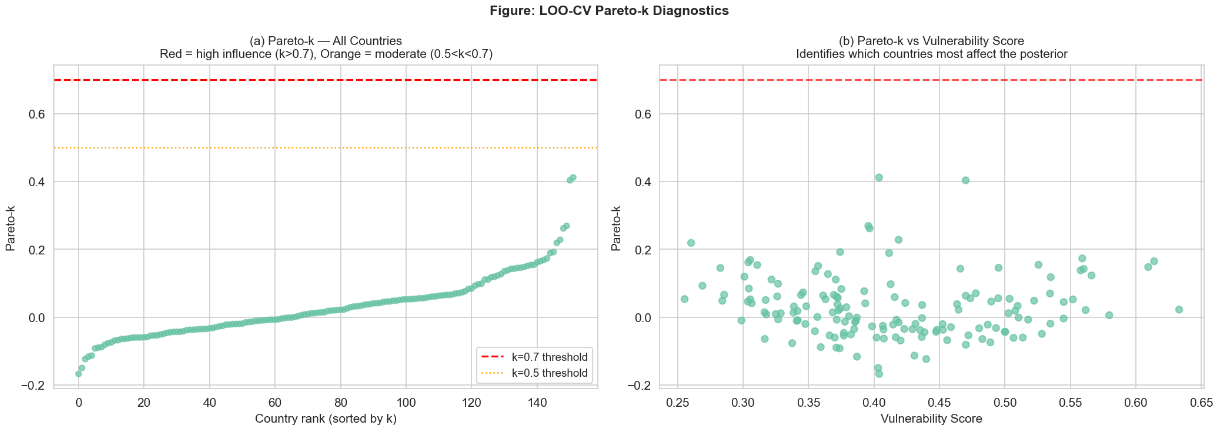

Figure 13: Pareto-k diagnostics. All 152 countries have $k < 0.5$ — the entire distribution sits below both the moderate (0.5) and high-influence (0.7) thresholds. No single country drives the posterior. The negative vulnerability coefficient is not an artefact of Samoa's extreme per-capita finance, India's extreme population, or any other outlier.

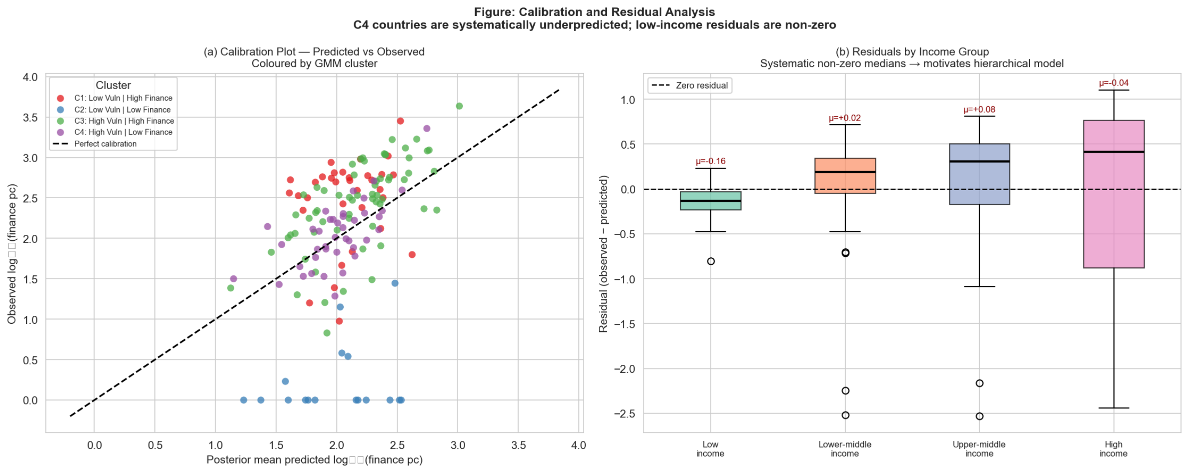

Figure 14: Calibration plot (predicted vs observed, coloured by cluster) and residuals by income group. Panel (a): C4 countries cluster below the perfect-calibration line — the model predicts more finance than they actually receive. Panel (b): mean residuals are −0.163 for low-income, +0.020 for lower-middle, +0.084 for upper-middle, −0.043 for high income. Non-zero within-group medians directly violate the pooled model's exchangeability assumption and motivate the hierarchical model.

C. Cluster-Level and Hierarchical Model Visualisations

This appendix presents the cluster-level and hierarchical model results referenced in Sections 5.7–5.9. Figure 15 shows the C4-only vs full-sample posterior comparison, supporting the structural-vs-within-cluster argument: the negative pooled coefficient is driven by between-cluster sorting rather than a within-C4 vulnerability gradient. Figure 16 shows all four income-group vulnerability coefficients under partial pooling, confirming the negative direction is consistent across the entire income distribution. Figure 17 and Table 6 together report the readiness decomposition: the figure shows the raw-readiness shrinkage that initially appeared to support institutional capacity as a clean mediator; the table shows that under orthogonalised readiness, the shrinkage reverses and the inseparability of vulnerability and institutional capacity becomes visible.

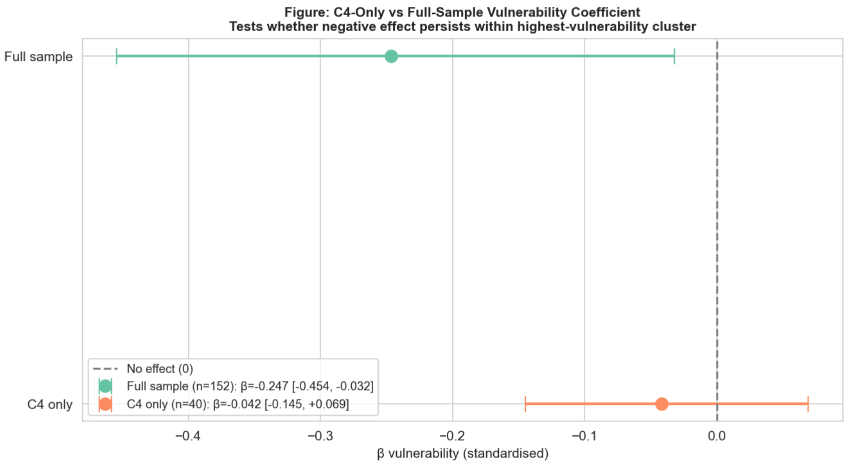

Figure 15: C4-only vs full-sample vulnerability coefficient. Within C4 ($n = 40$), $\beta = -0.042$ (94% HDI: $[-0.145, +0.069]$), $P(\beta < 0) = 77\%$. The HDI crosses zero, consistent with an underpowered within-cluster test. The direction aligns with the full sample but the primary mechanism is structural — the between-cluster sorting of institutionally weak, highly vulnerable countries — rather than a within-cluster gradient.

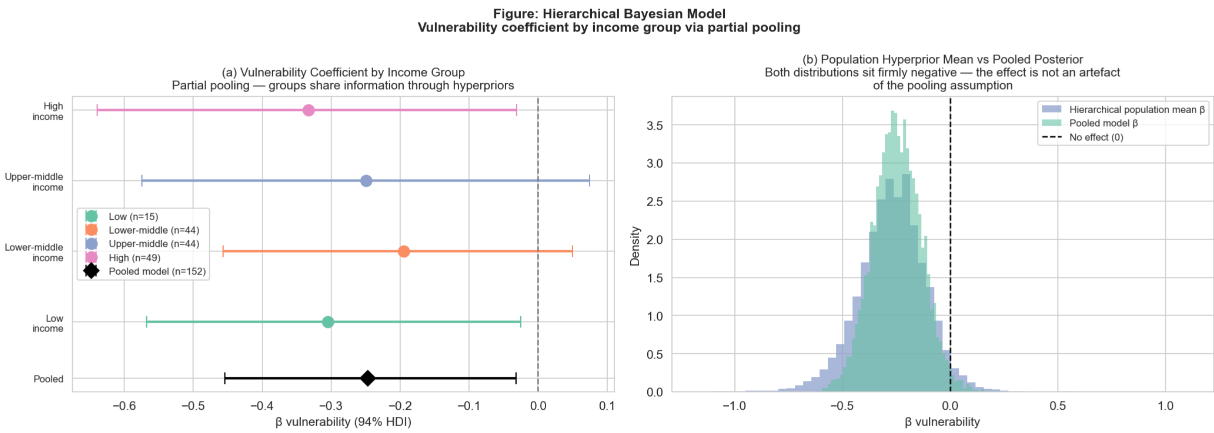

Figure 16: Hierarchical model results. Panel (a): all four income-group vulnerability coefficients are negative. Low income: $\beta = -0.305$, HDI excludes zero ($P(\beta < 0) = 98.6\%$). High income: $\beta = -0.333$, HDI excludes zero ($P(\beta < 0) = 98.7\%$). The negative effect is structural — it appears consistently across all income groups, not just in one. Cross-group SD $\sigma_\beta = 0.176$ indicates modest heterogeneity. Panel (b): the hierarchical population mean and the pooled posterior both sit firmly negative. Relaxing the exchangeability assumption does not change the headline finding.

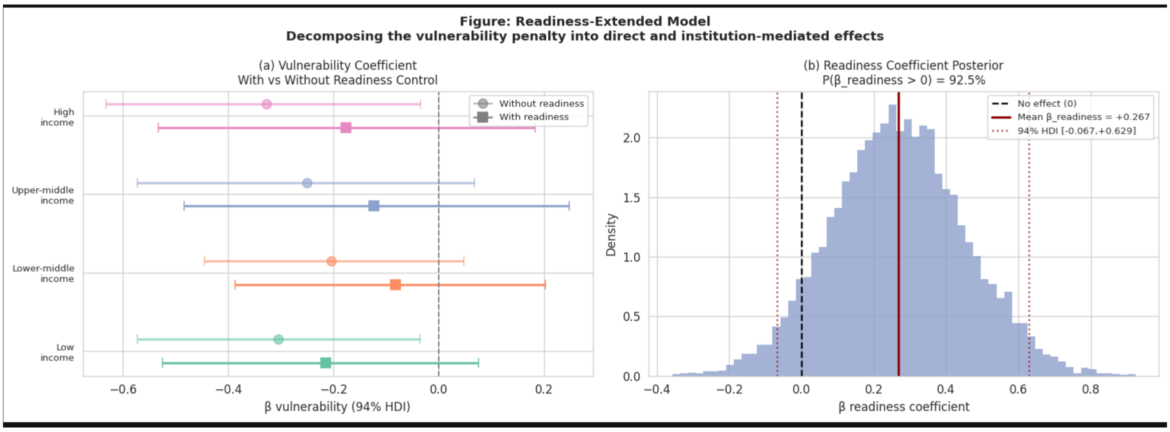

Figure 17: Readiness-extended hierarchical model: **raw** (non-orthogonalised) readiness control. Panel (a): forest plot of income-group vulnerability coefficients before (circles) and after (squares) the raw readiness adjustment. All four coefficients shift toward zero with shrinkage of approximately 46.5% on average. Panel (b): posterior density of the raw readiness coefficient, $\beta_{\text{readiness}} = +0.267$ (94% HDI: $[-0.067, +0.629]$), $P(\beta > 0) = 92.5\%$. The shrinkage pattern shown here is largely an artefact of the $\rho = -0.887$ correlation between vulnerability and readiness; under orthogonalised readiness (Table 6), the vulnerability coefficients move *away* from zero and the readiness coefficient is much smaller. The interpretation is that vulnerability and institutional capacity are empirically inseparable in this dataset, not that readiness mediates vulnerability per se.

Table 6: Vulnerability coefficient decomposition: raw readiness vs orthogonalised readiness. Hierarchical β is the coefficient without readiness control. Columns 3–4 report the shrinkage when raw readiness is added (positive = toward zero); columns 5–6 report the change when orthogonal readiness is used (negative = away from zero).

Income Group	**Hierarchical β**	**w/ Raw**	**Δ %**	**w/ Orth.**	**Δ %**
Low income	−0.305	−0.215	+29.6%	−0.450	−47.3%
Lower-middle income	−0.204	−0.083	+59.3%	−0.313	−53.7%
Upper-middle income	−0.251	−0.123	+50.9%	−0.356	−42.1%
High income	−0.328	−0.177	+46.0%	−0.415	−26.4%
Average			**+46.5%**		**−42.3%**